BENSON CYRIL NANA BOAKYE

Portfolio Website | +1 541-286-0930 | nanaboab@gmail.com | LinkedIn | Github

SUMMARY

Senior Data Scientist and Statistician with an M.S. in Statistics and 4+ years of experience architecting end-to-end ML systems, scalable data pipelines, and advanced statistical frameworks across healthcare, insurance, and research domains. Expert in designing and deploying production-grade predictive models using Python, R, SQL, and GCP, with deep fluency in XGBoost, survival analysis, NLP, and Bayesian inference. Proven ability to deliver executive-ready insights through Power BI and Tableau, lead cross-functional analytics initiatives, and translate ambiguous business problems into high-impact, data-driven solutions. Recognized for rigorous statistical methodology, MLOps best practices, and consistent delivery of measurable outcomes across complex, regulated environments.

CORE COMPETENCIES

- **Machine Learning & Advanced Modeling:** Supervised and unsupervised learning; gradient boosting (XGBoost, LightGBM), random forests, LASSO/Ridge, neural networks (TensorFlow/Keras), BERT, Cox proportional hazards, Kaplan-Meier, GLMs, Bayesian hierarchical models, SHAP explainability, PCA, t-SNE, k-NN; feature engineering, cross-validation, hyperparameter tuning (GridSearchCV), A/B testing, and experimental design.
- **Statistical Computing & Programming:** Python (pandas, NumPy, scikit-learn, TensorFlow, matplotlib, seaborn), R (dplyr, ggplot2, caret, survival), SQL, SAS; advanced statistical programming, simulation, reproducible analysis, and workflow automation.
- **MLOps, Data Engineering & Infrastructure:** MLflow experiment tracking, feature stores, CI/CD model deployment, ETL pipeline design, GCP (BigQuery, Cloud Storage), SQL Server, Oracle, MongoDB; data validation, governance, regulatory compliance, and workflow optimization. Power BI, Tableau, R-Shiny; executive dashboards, KPI frameworks, and stakeholder reporting.

PROFESSIONAL EXPERIENCE

Senior Data Scientist — **June 2025 – Present**
The Brooklyn Wellness Club — *Brooklyn, New York*

- Designed and deployed a multi-stage member health risk stratification system using XGBoost and Cox proportional hazards models on 15,000+ member records, powering personalized wellness program recommendations and driving a **22% lift** in 90-day retention.
- Engineered end-to-end Python ETL pipelines on GCP (BigQuery and Cloud Storage) ingesting data from wearables, EHR systems, appointment platforms, and nutrition logs, cutting manual data preparation by **60%** and enabling near-real-time analytics at scale.
- Fine-tuned a BERT-based NLP model on 8,000+ member survey responses for sentiment classification and theme extraction, delivering insights that informed a full program redesign resulting in a **17% increase** in Net Promoter Score.
- Designed and executed multivariate A/B tests across email, mobile push, and in-app wellness intervention campaigns; rigorous statistical analysis lifted conversion rates by **14%** and reduced churn-risk disengagement by **19%**.
- Developed a predictive member churn model using survival analysis with time-varying covariates, identifying at-risk members 60 days in advance at **81% recall**, enabling clinical staff to intervene proactively and prevent avoidable attrition.
- Established MLflow-tracked experiment infrastructure with automated model retraining, versioned feature stores, and CI/CD deployment pipelines, cutting model release cycles from 3 weeks to **4 days**.
- Built a 30+ KPI Power BI dashboard suite covering membership engagement, clinical program outcomes, and revenue performance, accelerating leadership decision cycles by **3x** over prior reporting workflows.

Senior Statistical Consultant — **September 2024 – June 2025**
Oregon State University, Department of Statistics — *Corvallis, OR*

- Directed statistical design and power analysis for 10+ biomedical studies including RCTs and longitudinal trials, optimizing sampling strategies to reduce required sample sizes by **20%** while fully preserving statistical power and study integrity.
- Engineered LASSO-penalized logistic regression pipelines for high-dimensional genomic data (1,000+ features), improving model AUROC by **∼70%** through rigorous cross-validation, hyperparameter tuning, and SHAP-driven feature selection.
- Directed RNA-seq transcriptomic analyses in Python and R, identifying statistically significant biomarkers and delivering reproducible, version-controlled workflows that contributed to two peer-reviewed publications.
- Served as lead statistical advisor to multidisciplinary research teams, applying GLM frameworks, survival models, and Bayesian inference to clinical biomarker studies spanning tuberculosis immunology, respiratory disease, and malnutrition.
- Conducted end-to-end biomarker analysis evaluating breath condensate pH, reflux events, and tracheal aspirate pepsin levels in SAS, producing validated statistical outputs that directly informed clinical interpretation and treatment protocols.

Graduate Teaching Assistant — **September 2023 – June 2025**
Oregon State University, Department of Statistics — *Corvallis, OR*

- Designed and delivered advanced statistical computing curriculum to **500+ graduate and undergraduate students**, covering ML pipelines (SVMs, ensemble methods, neural networks), dimensionality reduction (PCA, t-SNE), and applied inference using Python and R, driving a **15% improvement** in overall cohort performance across two academic years.
- Architected a suite of reusable, modular course materials built around end-to-end ML workflows, from data ingestion and feature engineering through model evaluation and SHAP interpretation, using version-controlled Jupyter notebooks and real-world datasets, reducing instructor preparation time by **40%** and accelerating industry-ready data science practice.
- Led graduate-level instruction on experimental design methodology including randomization schemes, blocking strategies, factorial and fractional factorial designs, and power analysis, equipping students to independently scope and execute rigorous studies.
- Mentored 20+ graduate students on applied research projects involving predictive modeling, survival analysis, and Bayesian inference, providing statistical guidance that directly contributed to three thesis defenses and two departmental research awards.

Senior Data & Research Analyst **October 2022 – August 2023**
Laboratory for Interdisciplinary Statistical Analysis, KNUST

- Developed Bayesian hierarchical models to quantify healthcare access disparities across demographic groups, identifying a **30% systemic gap** and producing findings that directly informed national-level public health policy recommendations.
- Engineered stratified sampling, post-stratification weighting, and multiple imputation pipelines in R, improving representation of underrepresented populations by **35%** and correcting **20% undercoverage** across 10 public datasets.
- Led statistical analysis across 7+ interdisciplinary COVID-19 impact studies in Python covering healthcare disparities, vaccine uptake, and telehealth access, with findings cited in two government-commissioned health equity reports.
- Designed and validated predictive models assessing socioeconomic determinants of health outcomes, applying mixed-effects regression and sensitivity analyses to ensure robustness across diverse population subgroups.
- Architected an automated data quality assurance framework in Python and R to validate incoming survey and administrative records across 10+ datasets, reducing data cleaning time by **45%** and ensuring analytical reproducibility.
- Developed and maintained a centralized research data warehouse integrating heterogeneous sources, including census records, clinical registries, and community health surveys, enabling cross-study querying and reducing dataset preparation time by **30%**.
- Conducted geospatial analysis and choropleth mapping of health outcome distributions using R (sf, ggplot2) across 12 administrative regions, producing visualizations directly embedded in policy briefs.
- Designed propensity score matching and inverse probability weighting (IPW) frameworks to control for confounding in observational health studies, strengthening causal inference validity across **4 peer-reviewed research outputs**.
- Presented analytical findings and model outputs to senior research leads and external stakeholders including NGO partners and government health officials, translating complex statistical results into actionable public health strategy.

Actuarial Data Analyst **September 2021 – December 2021**
National Insurance Trust

- Built TensorFlow-based policyholder behavior prediction models improving risk profiling accuracy by **24%**; deployed unsupervised anomaly detection pipelines in Python across **2TB** of claims data, reducing processing time by **20%**.
- Modeled reinsurance optimization strategies using Monte Carlo simulations for asset-liability management, contributing to investment decisions that achieved ∼**5% higher ROI** across client portfolios.
- Applied run-off triangle methods and actuarial pricing models to claims data and produced executive-facing Power BI and Tableau reports for senior stakeholder review and regulatory submission.
- Engineered a dynamic risk scoring system using gradient boosting and logistic regression on structured policyholder demographic and claims data, enabling actuarial teams to segment high-risk portfolios with **31% greater precision** and inform underwriting decisions across 50,000+ active policies.
- Designed and automated end-to-end actuarial reporting pipelines in Python and SQL, consolidating data from multiple legacy systems into a single validated reporting framework and reducing monthly close reporting time by **35%** while ensuring full regulatory compliance.

RELEVANT PROJECTS

ML - Based Prediction of IPF Progression | *Python, scikit-learn, XGBoost, SHAP, GridSearchCV*

- Trained and benchmarked 5 ML models on 1,129 proteomic features; Random Forest achieved **83.3% accuracy** and **AUC 0.984**. Applied SHAP explainability and GridSearchCV hyperparameter tuning to surface key biomarkers driving early clinical intervention and minimize overfitting across stratified folds.

Survival Analysis of Treatment Efficacy in Primary Biliary Cirrhosis | *R, SAS, Cox PH, Kaplan-Meier, AFT*

- Modeled Cyclosporin A treatment efficacy on PBC survival using Kaplan-Meier, AFT, and Cox PH models, improving prediction accuracy by **15%**; stratified analysis pinpointed elevated bilirubin and low albumin as primary mortality risk factors, enhancing clinical decision support.

HIV Prevention Resource Allocation: FSW Population Modeling | *R, Mixed-Effects, REML, NB/Poisson*

- Deployed zero-inflated mixed-effects models (negative binomial, Poisson) with REML estimation and country/region/year random effects ($p < 0.0001$, AIC 2863) to estimate FSW population distributions across Sub-Saharan Africa, directly informing WHO-aligned HIV intervention targeting and resource allocation.

Air Quality Trend Analysis Dashboard | *R, Shiny, ggplot2*

- Built an interactive R-Shiny application visualizing PM2.5 levels across U.S. and Indian cities with dynamic regional filtering and annotated COVID-19 lockdown events, enabling policy intervention impact analysis and temporal trend comparisons.

EDUCATION

Oregon State University **September 2023 – June 2025**
Master of Science in Statistics *CGPA: 3.74/4.0*

Kwame Nkrumah University of Science and Technology (KNUST) **August 2018 – September 2022**
Bachelor of Science in Actuarial Science *CGPA: 3.8/4.0*

WorldQuant University **August 2023 – September 2025**
Applied Data Analytics

RECENT PUBLICATION

- Fowotade, O.D., Makinde, J.O., Boakye, B.C.N., & Lasisi, S.O. (2025). Impact of Probiotics on Metabolic Interactions for the Prevention of Colorectal Cancer: A Comprehensive Network with Molecular Docking Studies. *Journal of Advances in Medical and Medical Research*, 37(6), 234–248. DOI link

CERTIFICATIONS & PROFESSIONAL MEMBERSHIP

- Member, American Statistical Association (2023) · SAS Programming Certification (2025) · Applied Data Analytics, WorldQuant University (2025).